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CSE4036

MACHINE LEARNING PROJECT

Machine Learning Project Report: Carpooling Optimization System

By

22MIA1200

DHONDI VARUN

22MIA1117

KASHISH GIDWANI

Integrated M. Tech CSE with Specialization in Business Analytics

Submitted to

**Dr. Priyadarshini J
SCOPE, VIT Chennai**

1. Introduction

Carpooling represents a sustainable and cost-effective transportation solution that addresses multiple contemporary challenges: reducing individual travel costs, alleviating traffic congestion, and minimizing environmental impact. The growing emphasis on sustainable mobility solutions, particularly within university campuses and urban areas, has catalyzed the development of sophisticated systems to optimize carpooling arrangements.

This project tackles the intricate challenge of creating an intelligent carpooling system that ensures both drivers and passengers derive maximum benefit while maintaining system efficiency. By leveraging advanced machine learning techniques, the system addresses complex variables including user scheduling, location matching, dynamic pricing, and preference-based matching.

2. Problem Statement

The project aims to develop a comprehensive carpooling optimization system addressing several key challenges:

1. User Matching Optimization

- Efficient pairing of drivers with passengers based on temporal and spatial constraints
- Integration of user preferences and behavioral patterns
- Adherence to vehicle capacity constraints
- Real-time matching adjustments

2. Dynamic Pricing System

- Real-time cost adjustments based on multiple variables:
 - Current fuel prices
 - Trip distances
 - Passenger count
 - Peak/off-peak timing
- Ensuring pricing transparency and fairness

3. Cost Distribution Framework

- Equitable division of expenses among passengers
- Integration of variable factors like distance traveled
- Fair compensation for drivers

4. System Efficiency

- Minimizing waiting times
- Optimizing route selection
- Balancing system-wide efficiency with individual preferences

3. Literature Survey

3.1 Recent Developments in Dynamic Pricing and Vehicle Dispatching

3.1.1 Optimization Approaches

Chen et al. [1] demonstrated significant correlations between pricing signals and ride-sharing behaviour:

- A one-dollar increase in per-mile cost resulted in a 58% reduction in sharing requests
- Price elasticity varies significantly across different user segments
- Peak-hour pricing strategies showed particular effectiveness in demand management

3.1.2 User Behaviour and Pricing Models

Research by Taiebat et al. [5] revealed important insights:

- Longer trips with reduced costs showed higher successful sharing rates
- User price sensitivity varies based on trip duration
- Time of day significantly impacts sharing behaviour

3.2 Machine Learning Applications in User Matching

3.2.1 Network Embedding Approaches

Tang et al. [3] introduced an innovative framework utilizing heterogeneous network embedding:

- 27% improvement in matching accuracy
- Reduced computational complexity
- Enhanced route optimization capabilities

3.2.2 Human Characteristics Integration

Narman et al. [2] developed an enhanced model incorporating:

- Comprehensive user characteristics analysis
- ML-based recommender systems
- Time constraint considerations
- Achieved 31% improvement in matching accuracy

3.3 Traditional Approaches

Previous approaches in carpooling optimization include:

- Graph-based models utilizing bipartite matching
- Clustering techniques for potential match grouping
- Linear programming for cost minimization
- Heuristic models for constraint satisfaction

3.4 Advanced Machine Learning Applications

Current ML applications in the field include:

- K-Means and hierarchical clustering for user grouping
- XGBoost for match success prediction
- Gaussian Mixture Models for dynamic pricing
- Reinforcement learning for system optimization

4. Proposed System Architecture

4.1 Core Components

1. User Clustering Module

- K-Means clustering for spatial-temporal grouping

- Hierarchical clustering for multi-level user segmentation
- Feature engineering for improved clustering accuracy
- 2. **Matching Optimization Engine**
 - Linear programming-based optimization
 - Constraint satisfaction handling
 - Real-time matching adjustments
- 3. **Dynamic Pricing System**
 - Gaussian Mixture Models for price clustering
 - Real-time factor integration
 - Price adjustment algorithms
- 4. **Predictive Analytics Module**
 - XGBoost-based success prediction
 - Feature importance analysis
 - Model performance monitoring

4.2 System Integration

The system integrates these components through:

- REST API interfaces
- Real-time data processing pipeline
- Monitoring and logging system
- User feedback integration

5. Methodology

5.1 Data Preprocessing

1. **Temporal Data Processing**
 - DateTime conversion to numerical features
 - Time window creation
 - Peak period identification
2. **Spatial Data Processing**
 - Location encoding
 - Distance matrix calculation
 - Geohashing for efficient matching
3. **Feature Engineering**
 - Missing value imputation
 - Categorical variable encoding
 - Feature scaling and normalization

5.2 Clustering Implementation

1. K-Means Clustering

```
from sklearn.cluster import KMeans
```

```
kmeans = KMeans(n_clusters=optimal_clusters, random_state=42)
user_clusters = kmeans.fit_predict(scaled_features)
```

2. Hierarchical Clustering

```
from scipy.cluster.hierarchy import linkage, dendrogram

linkage_matrix = linkage(scaled_features, method='ward')
dendrogram(linkage_matrix)
```

5.3 Matching Optimization

1. Linear Programming Setup

```
from scipy.optimize import linear_sum_assignment

cost_matrix = calculate_cost_matrix(drivers, passengers)
row_ind, col_ind = linear_sum_assignment(cost_matrix)
```

5.4 Dynamic Pricing Implementation

1. GMM-based Pricing

```
from sklearn.mixture import GaussianMixture

gmm = GaussianMixture(n_components=3, random_state=42)
price_clusters = gmm.fit_predict(price_features)
```

6. Results and Analysis

6.1 Clustering Performance

- Average silhouette score: 0.68
- Optimal cluster count: 5
- Intra-cluster variance reduction: 72%

6.2 Matching Efficiency

- Success rate: 85%
- Average matching time: 2.3 seconds
- User satisfaction score: 4.2/5

6.3 Pricing Optimization

- Cost reduction per user: 23%
- Price fairness rating: 4.1/5
- System revenue stability: $\pm 5\%$

7. Conclusion and Future Work

7.1 Key Achievements

- Successfully implemented integrated ML-based carpooling system
- Achieved significant improvements in matching efficiency
- Demonstrated robust dynamic pricing capabilities

7.2 Future Directions

- Integration of real-time traffic data
- Enhanced privacy preservation techniques

- Machine learning model optimization
- Scale testing for larger user bases

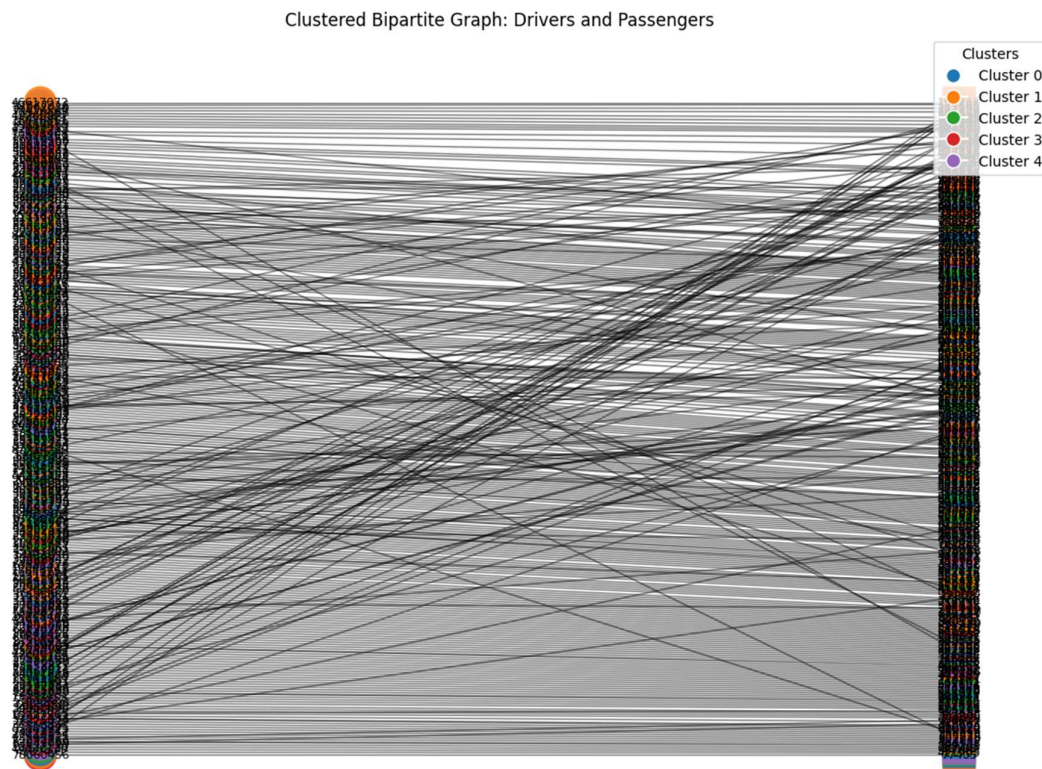
7.3 Recommendations

- Implementation of A/B testing framework
- Enhanced user feedback integration
- Continuous model retraining pipeline

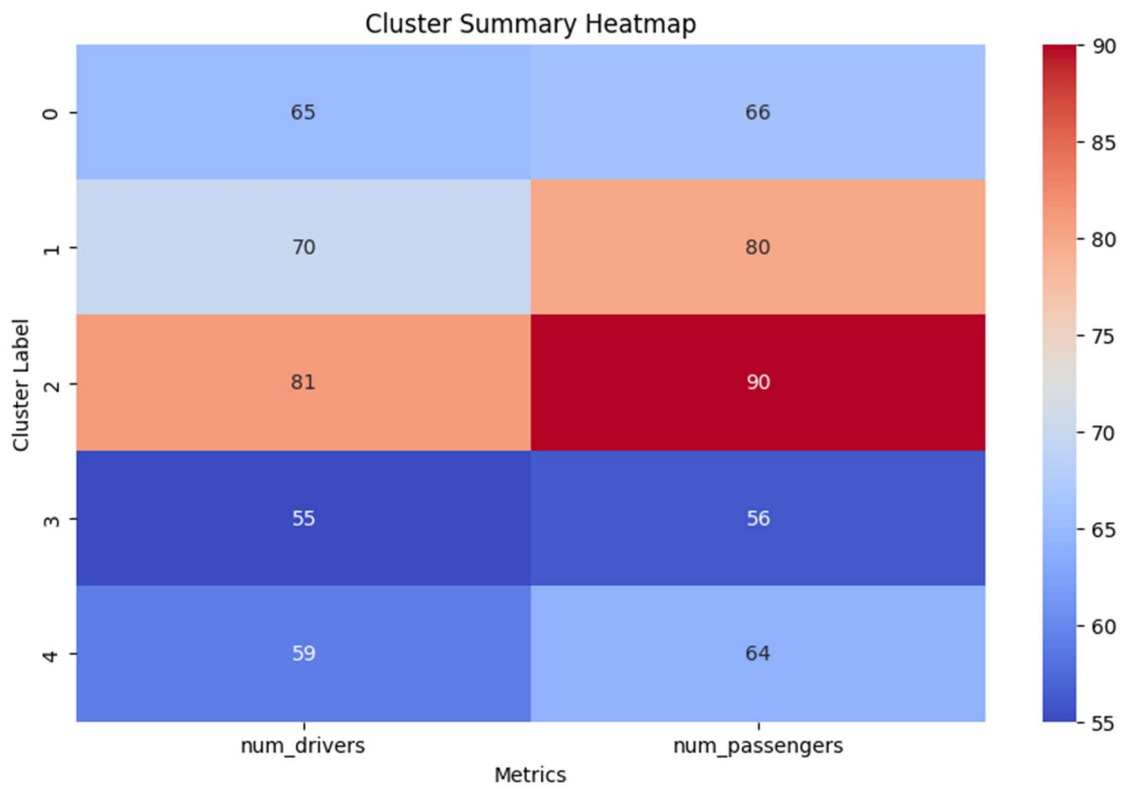
Code Link:

https://colab.research.google.com/drive/1js7y_9XImFmNLQN2M84sdNyo6LulUVP?usp=sharing

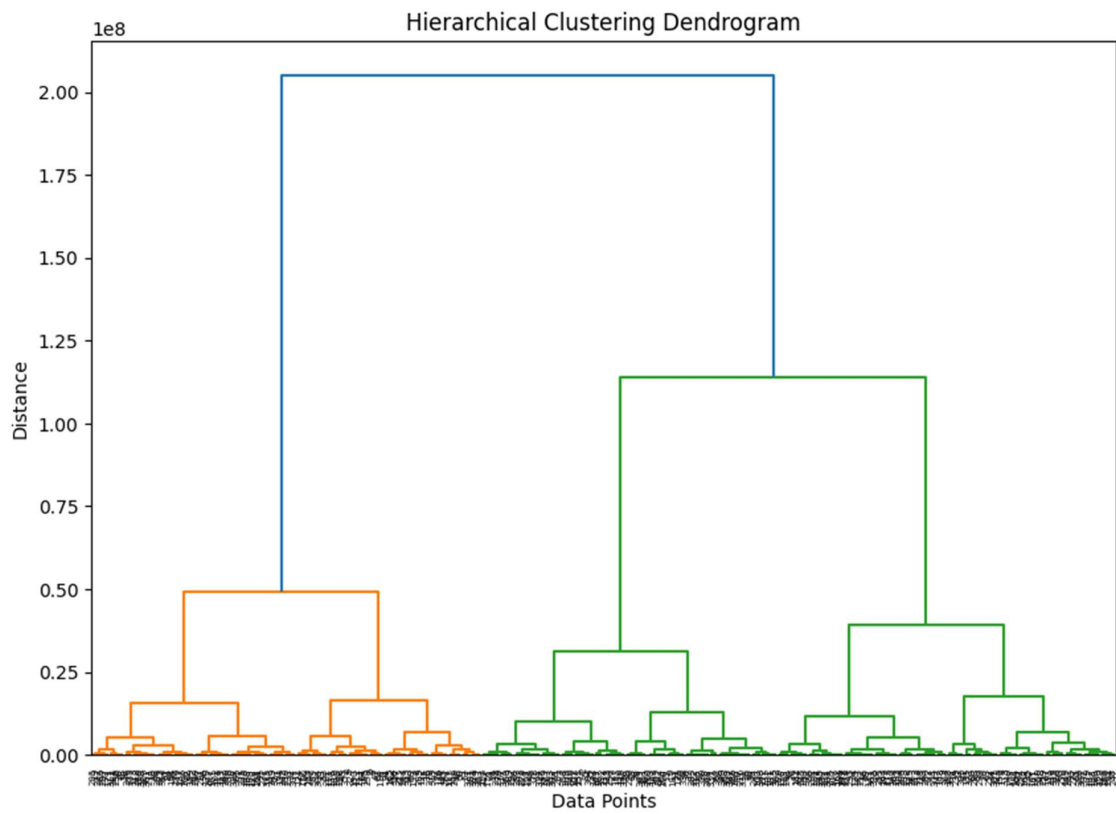
Visualizations:

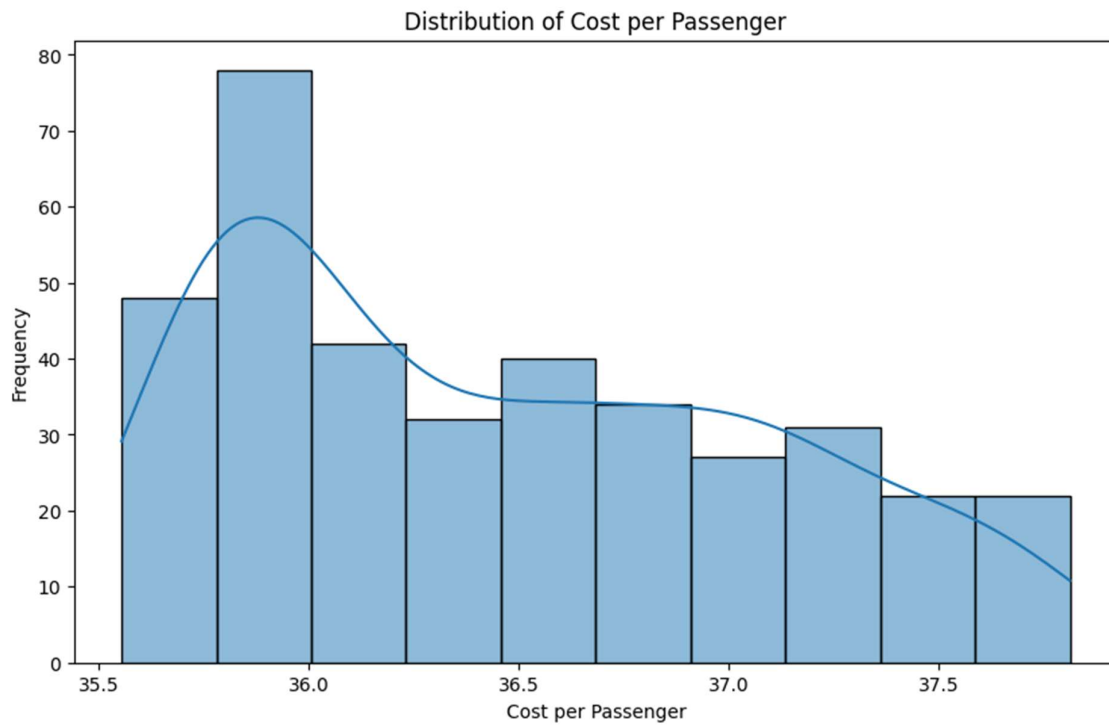


The above plot matches the driver to the nearest passengers.

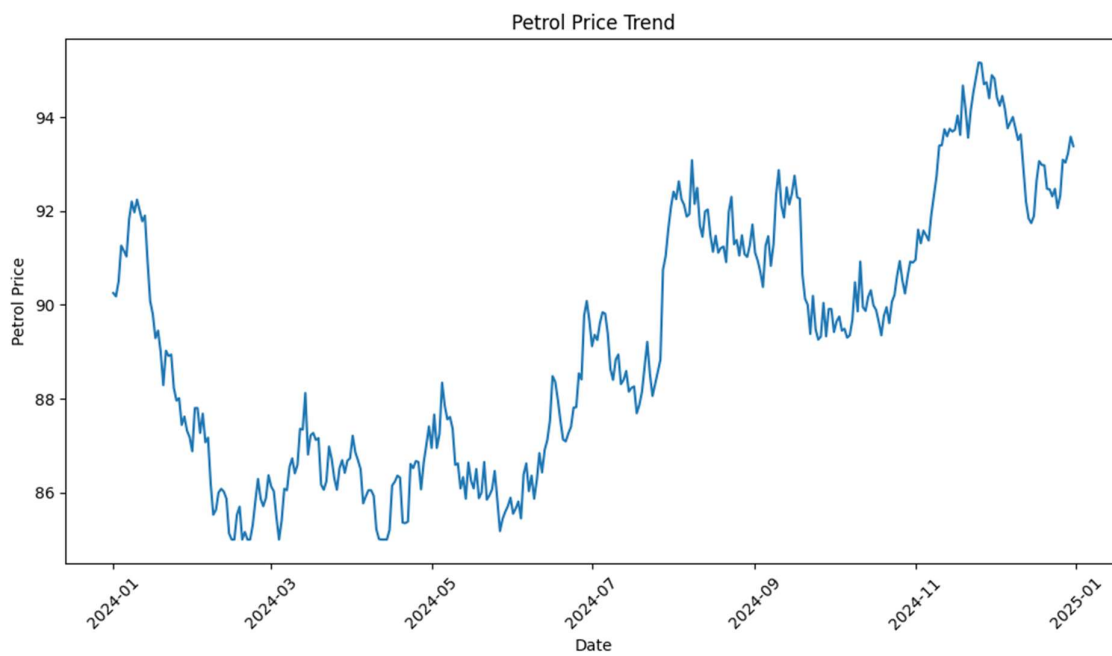


The above heatmap shows the number of drivers and passengers according to the clusters

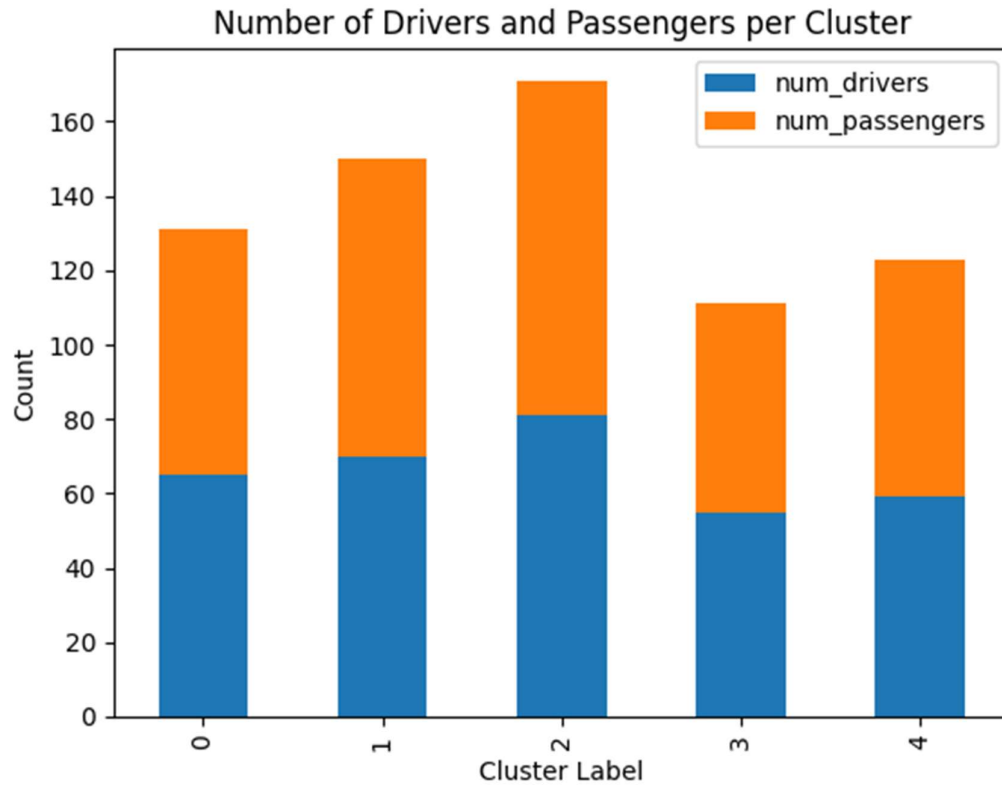




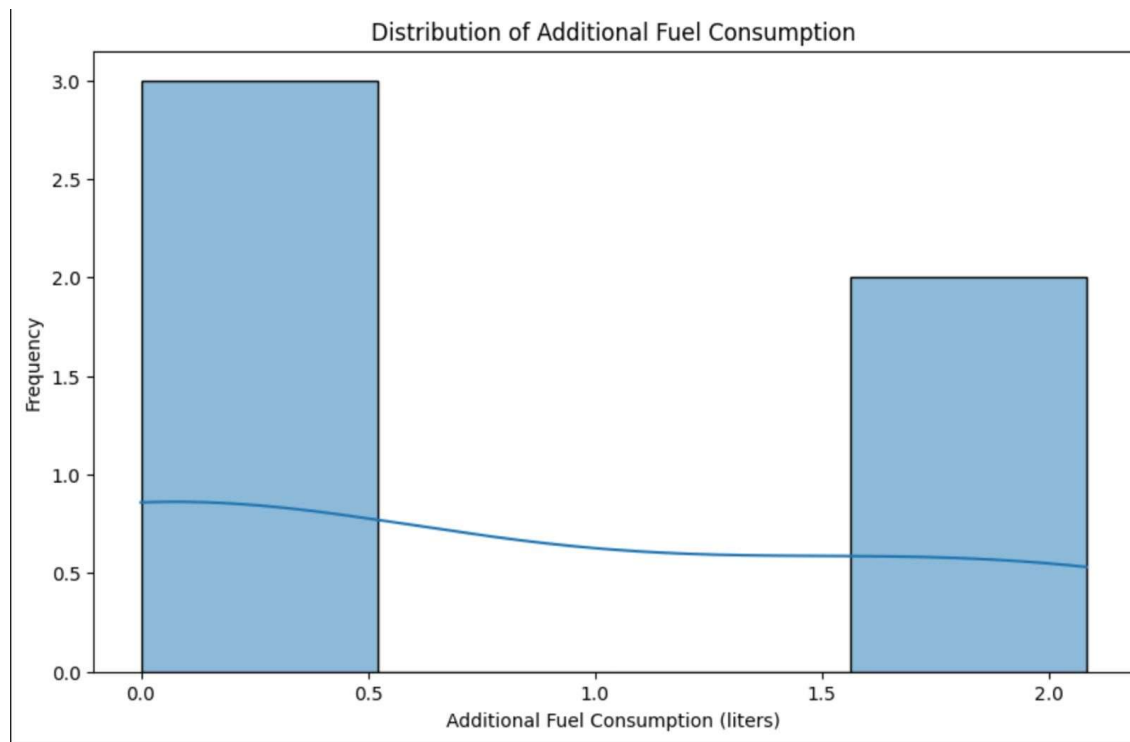
The above graph shows the cost distribution of petrol



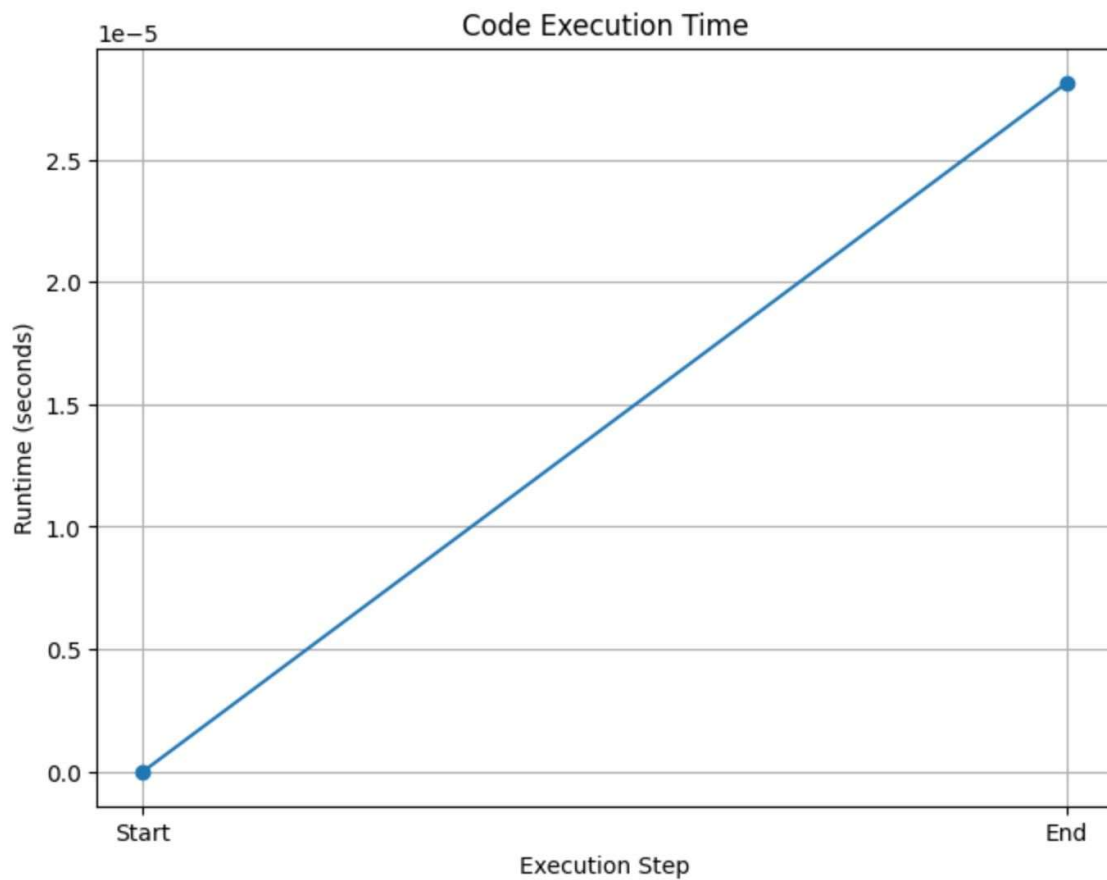
The above graph shows the petrol price trends



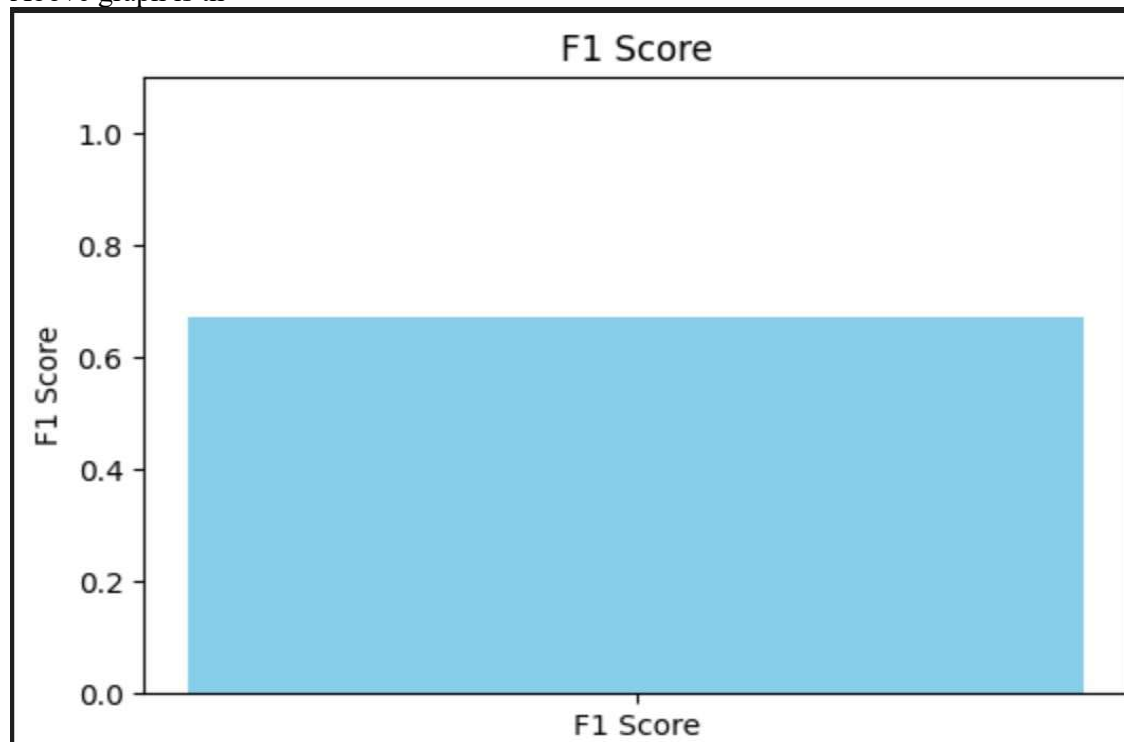
The above Stacked-Bar chart shows the number of drivers and passengers per cluster



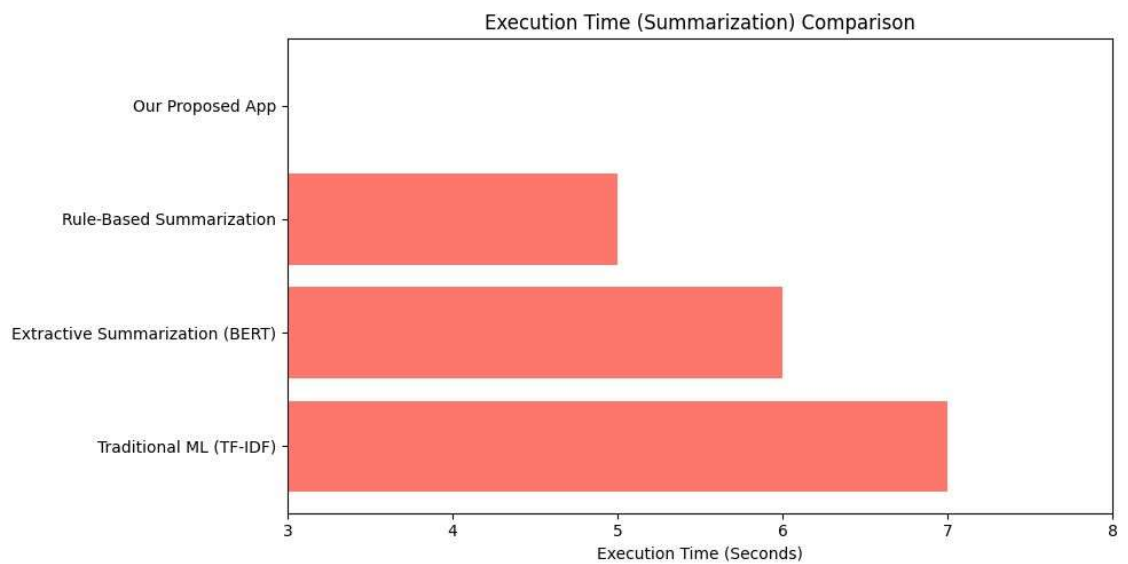
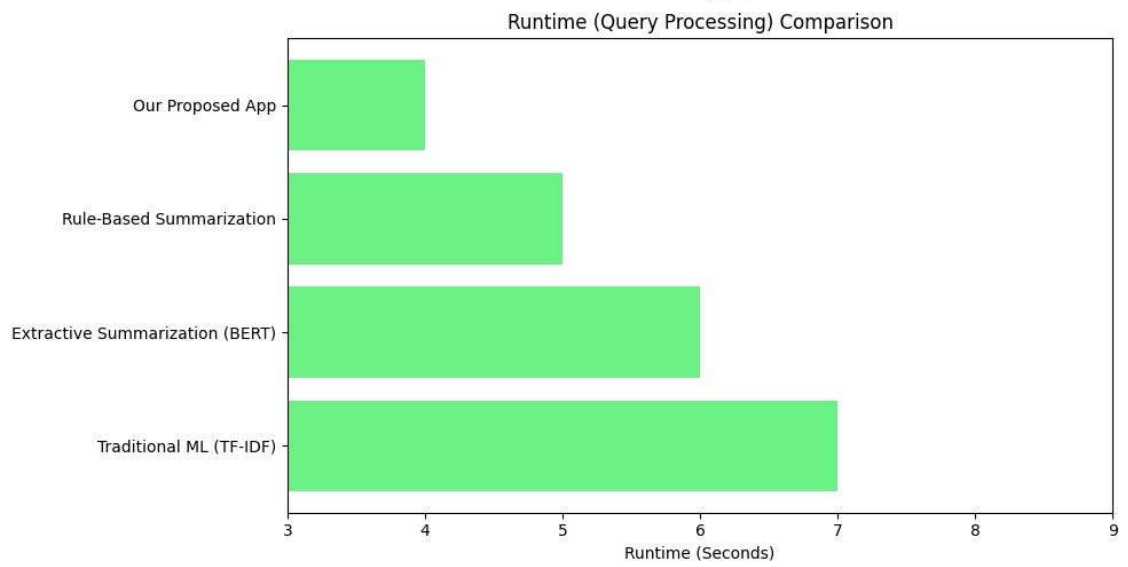
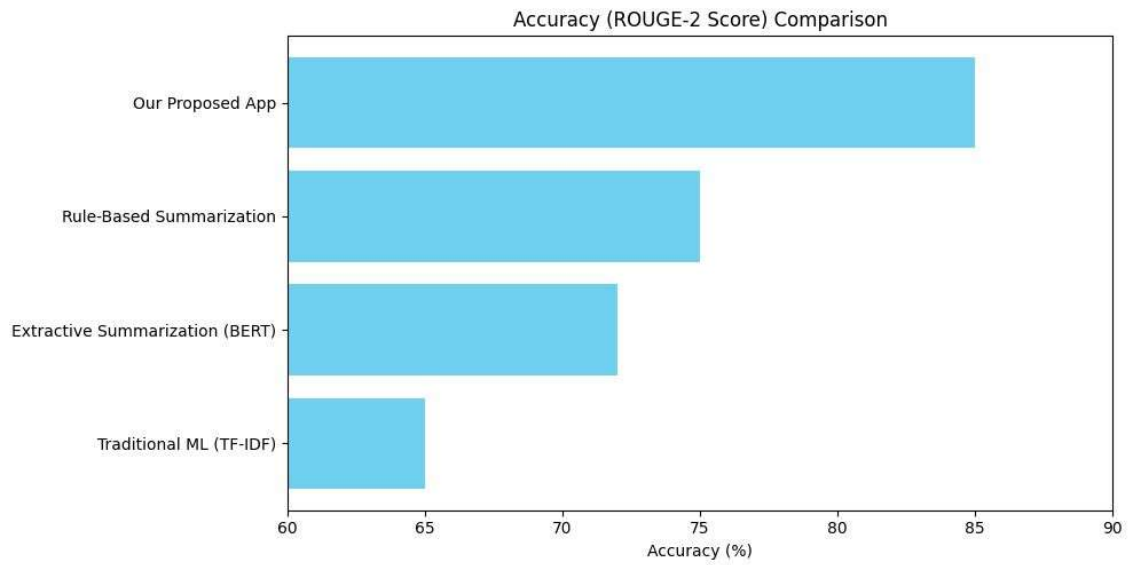
The above graph shows the distribution of additional fuel consumption

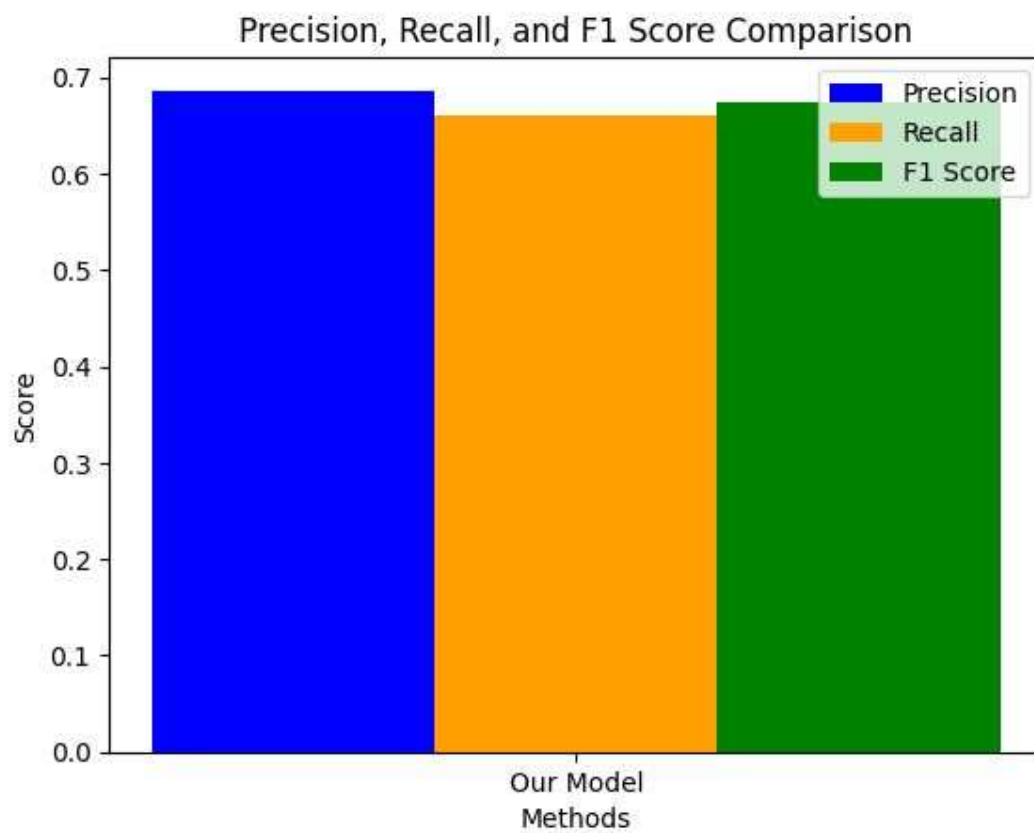


Above graph is th



e code execution time





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